INTERNSHIP PROJECT BRIEF / WEEK 02

EduPredict

Student Performance Prediction & Analysis

DOMAIN	PROJECT PERIOD	DELIVERY
AI Engineering	2–6 November 2026	Daily progress + Friday final

PROJECT LEVEL: Guided intermediate. A distinct real-world problem and new skills beyond Week 1; manageable within five working days.

01 | The real-world challenge

Educators can benefit from exploring patterns in learning-related data. Build a small, transparent classification prototype that demonstrates sound data preparation and evaluation, not a real-world decision system.

Intended user: A learner exploring educational data and model behavior; not an automated admissions or grading authority.

02 | Project objectives

- Select a public student dataset with a clear target, or use a provided clean sample.
- Explore and clean data; define a sensible performance-category target.
- Train one Logistic Regression or Decision Tree classifier.
- Evaluate on held-out data and provide a notebook or small Streamlit prediction interface.

03 | What you will learn

- Inspect tabular data and visualize distributions.
- Handle missing values and categorical features correctly.
- Separate training and test data and prevent target leakage.
- Interpret accuracy, confusion matrix and classification report; discuss limitations.

04 | Recommended tools and flexibility

- Python, pandas, matplotlib/seaborn, scikit-learn, Jupyter or Google Colab.
- Suggested data: UCI Student Performance dataset (archive.ics.uci.edu/dataset/320/student+performance).
- Use a notebook as the default interface; Streamlit is an optional alternative, not an extra requirement.
- No GPU, deep learning, API key or cloud deployment required.

Technology flexibility: You may use a familiar equivalent tool or framework if the mandatory outcome is achieved. Optional advanced tools, paid services and stretch features are not required.

05 | Your five-day project roadmap

Each day introduces a new concept and visible project progress. Submit daily evidence through the official Google Form; the complete project is due Friday.

Monday, 2 November | Understand dataset and prediction question

- Read dataset documentation and identify target and candidate input features.
- Load data with pandas; inspect columns, types, missing values and class balance.
- Define performance categories and explain why chosen inputs are available before the outcome.

Daily submission evidence: Notebook with dataset overview, target definition and initial plots.

Tuesday, 3 November | Prepare features without leakage

- Create train/test split before learning preprocessing statistics.
- Handle missing/categorical fields with a simple pipeline.
- Exclude final grades or other outcome-derived features when they leak the target.

Daily submission evidence: Documented preprocessing pipeline and leakage checklist.

Wednesday, 4 November | Train first classification model

- Train Logistic Regression or Decision Tree using the training split.
- Generate predictions on held-out test data.
- Record a simple baseline (e.g. majority class) for comparison.

Daily submission evidence: Model notebook, baseline and first evaluation outputs.

Thursday, 5 November | Evaluate and explain

- Produce confusion matrix, accuracy and precision/recall/F1 classification report.
- Investigate at least two wrong predictions or error patterns.
- Build a small prediction cell/form using the same preprocessing pipeline.

Daily submission evidence: Evaluation visuals, example predictions and limitations notes.

Friday, 6 November | Reproduce, document and demonstrate

- Restart and run notebook from top to bottom without hidden state.
- Document dataset source, assumptions, target leakage safeguards and limitations.
- Push notebook/source and record a narrated demonstration.

Daily submission evidence: GitHub repo, reproducible notebook, README, charts and narrated demo.

06 | Completion criteria: what must be ready by Friday

- Documented dataset, target definition and exploratory analysis.
- Train/test split and preprocessing pipeline with leakage safeguards.
- One trained classification model, baseline and meaningful evaluation metrics.
- Reproducible notebook or simple prediction UI and honest limitations.

Required final handover: Working project or complete design package, original source files, README/process notes, screenshots, narrated screen demonstration, and final submission through the official Google Form. Code projects require an accessible GitHub repository; design projects require editable source files and exports.

07 | Optional stretch challenges (not mandatory)

- Compare a second simple classifier.
- Cross-validation on the training set.
- Basic feature importance or coefficient interpretation with caveats.

Stretch ideas are for interns who finish the core work early. They are not required for successful completion.

08 | Daily submission and professional conduct

- Submit meaningful progress Monday–Friday through the official daily Google Form by 11:59 PM Pakistan time, with requested screenshots or links.
- Use Git regularly for coding projects and keep the final repository accessible. Designers should organize original editable files and exports.
- After completing the weekly project, post your work professionally on LinkedIn and tag the official Stack & Scale page. Saturday or Sunday posting is acceptable.
- You may learn from documentation, mentors and AI tools, but must understand and be able to explain submitted work.
- Never publish passwords, API keys, private customer data or unlicensed assets. Ask a mentor when blocked.

09 | Starter pack: start here

- Use UCI Student Performance dataset documentation to understand what each feature means.
- Avoid treating correlation as causation; predictions are educational only.
- Read scikit-learn documentation for Pipeline, train_test_split and classification_report.

Practical workflow: Read the daily goal → learn the smallest new concept → implement and test → capture evidence → commit/save source → submit progress.

10 | Friday handover checklist

- All mandatory features or assets are complete and demonstrable.
- Project runs correctly or exports open; source files are accessible and organized.
- README/process guide includes setup, use, sources, test notes and known limitations.
- Screenshots, final link/files and narrated demo are submitted through the Friday form.
- Be ready to explain a technical/design choice, a real challenge and what you learned.

11 | Mandatory Friday video demonstration

Every intern must submit a screen-recorded demonstration of the completed Week 2 project on Friday, 6 November 2026, through the official Google Form.

- Show the completed work and demonstrate all important mandatory features or design assets.
- Project-specific walkthrough: Explain the dataset and target, demonstrate preprocessing and one prediction, show confusion matrix and discuss one limitation or potential leakage risk.
- Explain at least one real problem, how you solved it and what you learned.
- Record a clear voice explanation and provide a viewable video file or link. If recording is inaccessible, contact your mentor beforehand to arrange an equivalent live demonstration.

12 | Responsible AI use and original work

- AI assistance for explanations, research, debugging, design exploration and learning is allowed.
- Do not submit copied or generated code, designs, models or workflows you cannot understand, test or explain.
- Mentors may ask you to demonstrate and defend project-specific decisions.
- Credit meaningful external resources or AI assistance in your README or design notes; protect private information.

Build Today. Scale Tomorrow. Week 2 focuses on new concepts, stronger problem-solving and independent delivery.